

NLP Basics

Duration: 3 hrs | Format: Lecture + Hands-on Demos

Session Overview

This session introduces Natural Language Processing (NLP). Learners will walk through the full NLP pipeline: from raw text cleaning and linguistic analysis, through classical text representation and word embeddings, to modern Transformer architectures and their real-world applications.

Learning Objectives

- Understand what NLP is and where it is used in everyday applications
- Distinguish the four levels of language analysis: Lexical, Syntactic, Semantic, Pragmatic
- Apply text preprocessing techniques: segmentation, tokenization, regex, stopwords removal, lemmatization
- Represent text as numbers using BoW, N-grams, TF-IDF, and word embeddings (Word2Vec, GloVe, FastText)
- Understand Transformer architecture — self-attention, multi-head attention, and encoder/decoder variants
- Apply pretrained Transformers to real NLP tasks using Hugging Face — classification, generation, translation & more

Session Agenda

Time	Topic	Coverage
PART 1 — Foundations (85 min)		
10 min	Introduction to NLP	What is NLP, real-world applications (search, chatbots, translation, spam filters), NLP pipeline overview
25 min	NLP Pipeline & Language Understanding	Lexical Analysis (POS tagging, morphology), Syntactic Analysis (dependency parsing, parse trees, chunking & phrase extraction), Semantic Analysis (word sense, semantic roles), Pragmatic Analysis (overview)
25 min	Text Preprocessing Basics	Sentence segmentation, tokenization, regex for text cleaning, noise removal, NLTK stopwords, stemming, lemmatization, normalization (lowercasing, punctuation, special chars)
15 min	Text Representation Techniques	Bag of Words, N-grams, TF-IDF, Word Embeddings intuition
10 min	Demo 1 — Pre-processing + Linguistic Analysis	Custom review dataset: sentence segment → tokenize → regex clean → POS tag → dependency parse → chunk → remove stopwords (NLTK) → lemmatize (NLTK) → one full preprocessing function, using spaCy + NLTK
Break (10 min)		
PART 2 — Embeddings, Transformers & Applications (85 min)		
25 min	Embeddings	One-hot vs dense vectors, Word2Vec (CBOW & Skip-gram, negative sampling), GloVe, FastText subword embeddings, cosine similarity, static vs contextual embeddings

30 min	Transformer Architecture	Self-attention (Query/Key/Value, scaled dot-product attention, worked numeric example), multi-head attention, positional encoding, encoder-only / decoder-only / encoder-decoder variants, Sentence Transformers
25 min	Demo 2 — Applications of Transformers	Live notebook walkthrough on application of transformers: classification, QA, NER, fill-mask, semantic similarity & clustering (encoder-only + Sentence Transformers), text generation (decoder-only), summarization, translation & zero-shot classification (encoder-decoder), and image classification/captioning (Vision Transformers)
5 min	Summary + Q&A	Key takeaways, recommended resources, next steps